
Problem 1

(a) Total points: 4

Recall definition of σ -algebra	1pt
Check inclusion of $\emptyset$ and Ω	1pt
Check inclusion of complement $E \setminus B$	1pt
Check inclusion of countable union	1pt

Model answer

First we note that $f^{-1}(\emptyset) = \emptyset \in \mathcal{F}$ and $f^{-1}(E) = \Omega \in \mathcal{F}$. So $\emptyset, E \in \mathcal{H}$.

Next, let $B \in \mathcal{H}$. Then

$$f^{-1}(E \setminus B) = \Omega \setminus f^{-1}(B) \in \mathcal{F},$$

since by definition $f^{-1}(B) \in \mathcal{F}$. So $E \setminus B \in \mathcal{H}$.

Finally, if $(B_i)_{i \in \mathbb{N}}$ is a sequence of sets in $\mathcal{H}$, then

$$f^{-1}\left(\bigcup_{i=1}^{\infty} B_i\right) = \bigcup_{i=1}^{\infty} f^{-1}(B_i) \in \mathcal{F},$$

which shows that $\bigcup_{i=1}^{\infty} B_i \in \mathcal{H}$, completing the proof that $\mathcal{H}$ is a σ -algebra.

(b) Total points: 3

Recall inclusion property of σ -algebras	1pt
Note that $\mathcal{A} \subseteq \mathcal{H}$ and thus $\mathcal{G} \subseteq$	1pt
Conclude	1 pt

Model answer By construction $\mathcal{A} \subseteq \mathcal{H}$. It therefore follows from Lemma 2.5 of the Lecture Notes that $\mathcal{G} = \sigma(\mathcal{A}) \subseteq \mathcal{H}$. But this then implies that $f^{-1}(B) \in \mathcal{F}$ for each $B \in \mathcal{G}$ which means that f is $(\mathcal{F}, \mathcal{G})$ -measurable.

Problem 2

(a) Total points: 2

Show the property holds for simple functions	1pt
Use the definition and non-negativity to conclude	1pt

Model answer

Let $g = \sum_{i=1}^N a_i \mathbb{1}_{A_i}$ is a simple function. Then $g \mathbb{1}_B = \sum_{i=1}^N a_i \mathbb{1}_{A_i \cap B}$ is also a simple function and thus

$$\int_B g \, d\mu = \int_{\Omega} g \mathbb{1}_B \, d\mu = \sum_{i=1}^N a_i \mu(A_i \cap B) \leq \mu(B) \sum_{i=1}^N a_i \mu(A_i) = 0.$$

Now let f be a non-negative function and $g \leq f$ be a simple function. Then $g \mathbb{1}_B \leq f \mathbb{1}_B$ and thus by Definition 4.7 and the calculation above

$$\int_B f \, d\mu = \int_{\Omega} f \mathbb{1}_B \, d\mu \geq \int_{\Omega} g \mathbb{1}_B \, d\mu = 0,$$

which implies the result.

(b) Total points: 1

Use monotonicity of supremum to conclude	1pt
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Model answer

Suppose $f \leq g$ are non-negative functions and observe that if h is a simple function such that $h \leq f$ then also $h \leq g$. Therefore we get

$$\int_{\Omega} f \, d\mu = \sup_{h \leq f} \left\{ \int_{\Omega} h \, d\mu \right\} \leq \sup_{h \leq g} \left\{ \int_{\Omega} h \, d\mu \right\} = \int_{\Omega} g \, d\mu.$$

(c) Total points: 3

Show the property holds for simple functions	1pt
Use previous result and homogeneity of supremum to conclude	2pt

Model answer

Suppose that h is a simple function. Then αh is also simple and it immediately follows that $\int_{\Omega} (\alpha h) \, d\mu = \alpha \int_{\Omega} h \, d\mu$.

Now let f be non-negative. Then $h \leq f \iff \alpha h \leq \alpha f$ and $h \leq \alpha f \iff \alpha^{-1} h \leq f$. Thus by Definition 4.7 we have

$$\begin{aligned} \alpha \int_{\Omega} f \, d\mu &= \alpha \sup_{h \leq f} \left\{ \int_{\Omega} h \, d\mu \right\} \\ &= \sup_{h \leq f} \alpha \left\{ \int_{\Omega} h \, d\mu \right\} \\ &= \sup_{h \leq f} \left\{ \int_{\Omega} (\alpha h) \, d\mu \right\} \\ &= \sup_{\alpha^{-1} h \leq f} \left\{ \int_{\Omega} h \, d\mu \right\} \\ &= \sup_{h \leq \alpha f} \left\{ \int_{\Omega} (\alpha h) \, d\mu \right\} = \int_{\Omega} (\alpha f) \, d\mu. \end{aligned}$$

Problem 8.3

Total points: 5

$\Rightarrow$

Write $Y_n = |X_n - X|$ and use definition of convergence in probability to conclude this converges in distribution to 0 function 1pt

Use alternative definition of convergence in distribution (Lemma 8.2) and the fact that $\varepsilon > 0$ is a continuity point of the 0 function to conclude 1pt

$\Leftarrow$

Use Lemma 8.2 together with definition of convergence in probability to identify statement to prove 1pt

Note that 0 is the only non-continuity point and the result is trivial for $t < 0$ 1 pt

Use the assumption to conclude the result for $t > 0$ 1 pt

Model answer

The main idea is to use the equivalent version of convergence in distribution.

Suppose that $X_n \xrightarrow{\mathbb{P}} X$ and define $Y_n = |X_n - X|$. We need to show that $\mathbb{P}(Y_n > \varepsilon) \rightarrow 0$ holds for any $\varepsilon > 0$.

First recall that $X_n \xrightarrow{\mathbb{P}} X$ is defined as weak convergence of Y_n to the constant zero random variable. By Lemma 8.2 this is equivalent to

$$\lim_{n \rightarrow \infty} \mathbb{P}(Y_n \leq t) = \mathbb{P}(0 \leq t),$$

for all continuity points of the function $\omega \mapsto 0$. We now note that any $\varepsilon > 0$ is a continuity point of this function. Hence, we get

$$\lim_{n \rightarrow \infty} \mathbb{P}(Y_n > \varepsilon) = 1 - \lim_{n \rightarrow \infty} \mathbb{P}(Y_n \leq \varepsilon) = 1 - \mathbb{P}(0 \leq \varepsilon) = 0$$

Now we prove the other implication. So suppose that $\mathbb{P}(Y_n > \varepsilon) \rightarrow 0$ holds for any $\varepsilon > 0$. We then have to prove that $(Y_n)_\# \mathbb{P} \Rightarrow 0_\# \mathbb{P}$. Due to Lemma 8.2 it is enough to show that

$$\lim_{n \rightarrow \infty} \mathbb{P}(Y_n \leq t) = \mathbb{P}(0 \leq t) = \mathbb{1}_{t \geq 0},$$

holds for all continuity points t of the function $\omega \mapsto 0$. Notice that the only non-continuity point is 0. Moreover, for all $t < 0$ we have that $\mathbb{P}(Y_n \leq t) = 0$ since $Y_n \geq 0$ almost every-where. Finally, for all $t > 0$ we have

$$\lim_{n \rightarrow \infty} \mathbb{P}(Y_n \leq t) = 1 - \lim_{n \rightarrow \infty} \mathbb{P}(Y_n > t) = 1 = \mathbb{P}(0 \leq t).$$

Problem 11.2

(a) Total points: 2

Note that the events $\{f \geq g\}, \{f < g\}$ are $\mathcal{H}$ -measurable 1pt

Split the integral over these two sets and use assumption to conclude 1pt

First note that since $f - g$ is $\mathcal{H}$ -measurable, we have that $\{f \geq g\}, \{f < g\} \in \mathcal{H}$. We then write

$$\int_{\Omega} \|f - g\| d\mathbb{P} = \int_{f \geq g} (f - g) d\mathbb{P} - \int_{f < g} (f - g) d\mathbb{P}.$$

Since $\int_B f d\mathbb{P} = \int_B g d\mathbb{P}$ holds for all $B \in \mathcal{H}$ both integrals on the right hand side are zero.

(b) Total points: 2

Use definition to conclude the integral are the same.	1pt
Use assumption and part a) to conclude	1pt

Model answer

By definition we have that

$$\int_B \mathbb{E}[X|\mathcal{H}] \, d\mathbb{P} = \int_B X \, d\mathbb{P},$$

holds for all $B \in \mathcal{H}$.

Since by assumption both $\mathbb{E}[X|\mathcal{H}]$ and X are $\mathcal{H}$ -measurable the result follows from part a).

(c) Total points: 2

Observe that $\mathbb{E}[X \mathcal{H}] + \mathbb{E}[Y \mathcal{H}]$ is $\mathcal{H}$ -measurable.	1pt
Show equality of integrals for all $B \in \mathcal{H}$ and use a) to conclude	1pt

Model answer

Similarly to the previous point, we first note that since $\mathbb{E}[X|\mathcal{H}]$ and $\mathbb{E}[Y|\mathcal{H}]$ are $\mathcal{H}$ -measurable so is $\mathbb{E}[X|\mathcal{H}] + \mathbb{E}[Y|\mathcal{H}]$. The result then follows from part a) because

$$\begin{aligned} \int_B \mathbb{E}[X|\mathcal{H}] + \mathbb{E}[Y|\mathcal{H}] \, d\mathbb{P} &= \int_B \mathbb{E}[X|\mathcal{H}] \, d\mathbb{P} + \int_B \mathbb{E}[Y|\mathcal{H}] \, d\mathbb{P} \\ &= \int_B X \, d\mathbb{P} + \int_B Y \, d\mathbb{P} = \int_B X + Y \, d\mathbb{P}. \end{aligned}$$

(d) Total points: 3

Use monotonicity of integration to get inequality for integrals of $\mathbb{E}[X \mathcal{H}]$ and $\mathbb{E}[Y \mathcal{H}]$ for each B	1pt
Introduce event $\{\mathbb{E}[X \mathcal{H}] > \mathbb{E}[Y \mathcal{H}]\}$ in $\mathcal{H}$	1pt
Prove that this event has measure zero by contradiction	1 pt

Model answer

First we observe that for any $B \in \mathcal{H}$

$$\int_B \mathbb{E}[X|\mathcal{H}] \, d\mathbb{P} = \int_B X \, d\mathbb{P} \leq \int_B Y \, d\mathbb{P} = \int_B \mathbb{E}[Y|\mathcal{H}] \, d\mathbb{P}.$$

Now consider the event $A := \{\mathbb{E}[X|\mathcal{H}] > \mathbb{E}[Y|\mathcal{H}]\} \in \mathcal{H}$. If this event has non-zero measure then it would follow that

$$\int_A \mathbb{E}[X|\mathcal{H}] \, d\mathbb{P} > \int_A \mathbb{E}[Y|\mathcal{H}] \, d\mathbb{P},$$

which is a contradiction. Hence we conclude that $\mathbb{E}[X|\mathcal{H}] \leq \mathbb{E}[Y|\mathcal{H}]$ holds $\mathbb{P}$ -almost everywhere.